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Romain Raffin (LIB Laboratory, Université de Bourgogne)
Gilles Rougeron & Jeremie Le Garrec (LSI Department, CEA LIST)

Subject: Application for PhD Position – Automatic construction of a digital model from real-world scenes (PC5)

Dear Romain Raffin, Gilles Rougeron, and Jeremie Le Garrec,

I am writing to express my eager interest in the PC5 PhD position on the "Automatic construction of a digital model from real-world scenes," hosted by the LIB laboratory and the LSI department at CEA LIST. Given my extensive work in Scientific Machine Learning (SciML) and 3D web visualizations, I am highly motivated to tackle the challenge of extracting geometry, kinematics, and mechanical properties from videos and photos to build robust, interactive digital models.

Bridging the gap between raw 2D visual data and interactive 3D digital replicas is a core theme in my engineering portfolio. For my "Live Control Room" project, I engineered a high-performance Next.js and Three.js 3D dashboard that seamlessly visualizes dynamic physical system states. This hands-on experience dealing with camera matrices, scene graphs, and web-based rendering engines provides me with the foundational skills necessary to translate raw image captures into coherent spatial and geometric representations.

Furthermore, beyond simple geometric reconstruction, I am deeply interested in your objective to deduce complex physical properties directly from observational data. By combining my rendering experience with my work in Multi-Fidelity Scientific Machine Learning, I have learned how to embed physical laws directly into deep learning architectures. I am uniquely positioned to apply deep learning not just for computer vision tasks, but to deduce complex mechanical properties from mere visual observations, automating the creation of high-fidelity functional twins. I thrive in interdisciplinary environments and look forward to validating these methodologies across the diverse industrial case studies provided by the EDT program. Please find attached my resume and academic transcripts.

Sincerely,

Ismail Aissaoui